

Deterministic algorithms over represented finite fields: consequences of prime-field factorization

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Abstract

We spell out uniform deterministic reductions for complete dense polynomial factorization and prescribed-degree irreducible construction over $K = \mathbf{F}_p[t]/(h)$, where p is prime and h is a supplied monic irreducible polynomial. Their only oracle is prime-field polynomial factorization. A trace-coordinate implementation and an independent prime-Frobenius implementation handle multiplicities and inseparability. Substituting the prime-field theorem of OpenAI gives polynomial bit complexity in the dense input and requested output sizes. The reductions are established machinery of Berlekamp and Shoup; the degree-multiplication construction also appears in Rai. This note records consequences, representation details and an audit, not a new prime-field algorithm or a priority claim. The upstream theorem requires the general cyclotomic Hecke zero-free theorem of a separate companion, beyond the narrower formalized quasi-Riemann theorem.

1 Problem, provenance and exact model

Represent p in binary, put $L = \lceil \log_2 p \rceil$, and supply $h \in \mathbf{F}_p[T]$ monic of degree $m \geq 1$. Primality of p and irreducibility of h are promises, or may be checked deterministically. The field $K = \mathbf{F}_p[T]/(h)$ has power basis $1, t, \dots, t^{m-1}$; every coordinate is a canonical residue with L bits. A nonzero polynomial $f \in K[X]$ of degree n is a dense list. Complete factorization returns its leading coefficient c and distinct monic irreducibles g_i with positive binary multiplicities e_i , such that $f = c \prod_i g_i^{e_i}$. Constants have an empty list. Zero has no such finite unique factorization and is rejected. The field and coefficient input uses $O((n+2)mL)$ bits; output coefficients including the leading coefficient use $O((n+1)mL)$ and multiplicities $O(n \log(n+1))$ bits.

For construction, a positive degree d may be supplied in binary, but the requested dense output has $(d+1)m$ coefficient slots. Polynomial in d is polynomial in this output size, and is not polynomial in $\log d$. No practical efficiency is inferred from a fixed polynomial exponent. A dense-input convention cannot be inferred from a field cardinality alone; the supplied defining polynomial is part of the input.

Berlekamp's reduction from general finite fields to split prime-field polynomials is already public [1]. Shoup's construction reduction is public in Theorems 3.1 and 4.1 of [2]. The particular degree- md lift below also occurs in Algorithm 2 and Theorem 4.1 of Rai [3]. The newly supplied input is OpenAI's prime-field theorem [4]. We attribute that breakthrough to its source. Consolidating its implications does not independently solve the base deterministic factorization problem or invent these reductions.

2 Explicit reduction for complete factorization

Let $P(u, L)$ be a nondecreasing bit-cost bound for a deterministic complete factorization algorithm PF over $\mathbf{F}_p$ on degree at most u .

Theorem 1 (Established reduction with explicit oracle accounting). *There is a uniform deterministic reduction for complete factorization in the above representation with polynomial nonoracle overhead in n, m, L . The trace version uses at most mn calls to PF, each of degree at most n , with sum of queried degrees at most mn^2 . An alternative uses at most n calls and sum of degrees at most n^2 . Neither version enumerates the field or uses an integer-factorization or primitive-root oracle.*

2.1 Fixed algebras and trace separation

First suppose f is monic squarefree of degree u . If it has s irreducible components, the Chinese remainder theorem gives

$$A = K[X]/(f) \simeq \prod_{a=1}^s \mathbf{F}_{p^{mu_a}}, \quad \sum_a u_a = u, \quad s \leq u.$$

With $q = p^m$, q -Frobenius is K -linear. Its fixed algebra $B_q = \ker(\text{Frob}_q - I)$ is K^s . Form its $u \times u$ matrix and compute a kernel basis $b_1, \dots, b_s$ by exact elimination. For $0 \leq j < m$ form

$$c_{ij} = \sum_{k=0}^{m-1} (t^j b_i)^{p^k} \in A.$$

Each component of b_i belongs to K , so each component of c_{ij} is $\text{Tr}_{K/\mathbf{F}_p}(t^j b_i(a)) \in \mathbf{F}_p$. This is a quotient-algebra power computation, not a coefficientwise trace on arbitrary elements of A .

The trace polynomial $\sum_{k=0}^{m-1} Y^{p^k}$ is nonzero and has degree $p^{m-1} < p^m$, so trace is nonzero, including when $p \mid m$. For every $\delta \neq 0$, multiplication by δ is bijective; therefore $y \mapsto \text{Tr}(\delta y)$ is nonzero. Some power-basis coordinate $y = t^j$ detects it. Two different coordinate maps on $B_q = K^s$ differ on some b_i ; hence some c_{ij} separates every pair of components. Using just $\text{Tr}(b_i)$ would not justify this claim: $\text{Tr}(1)$ can vanish.

If c has component values $a_1, \dots, a_s \in \mathbf{F}_p$, its monic minimal polynomial, over either K or $\mathbf{F}_p$, is exactly

$$\mu_c(Z) = \prod_{a \in \{a_1, \dots, a_s\}} (Z - a).$$

A polynomial annihilates c precisely when it vanishes at each coordinate; this proves the formula, squarefreeness and degree at most s . Find μ_c by the first dependence among $1, c, \dots, c^s$ in the u -dimensional K -space A . Its coefficients lie in the canonical $\mathbf{F}_p$ and can be converted after checking the other basis coordinates vanish. Factor μ_c with PF. The returned factors are distinct linear polynomials and give its occurring values. Refine each current factor block g by $\gcd(g, c - a)$ for those values a . These gcds partition its component support. After all ms coordinates no block contains two components, so every block is irreducible.

2.2 Independent prime-Frobenius variant

The p -power map is $\mathbf{F}_p$ -linear, generally not K -linear. On the um -dimensional space with basis $t^j X^i$, form $\text{Frob}_p - I$ as an $um \times um$ matrix over $\mathbf{F}_p$. Its kernel B_p is $\mathbf{F}_p^s$, because the p -fixed elements of every finite field component are exactly $\mathbf{F}_p$. A basis of B_p separates all coordinate

pairs. For each of its s basis vectors compute the first power relation over $\mathbf{F}_p$ and apply the same minimal-polynomial factorization and gcd refinements. This proves correctness independently of the trace pairing and uses only s prime calls.

For one squarefree input, the trace version uses at most ms^3 gcds and the direct version at most s^3 . A first-dependence system has u rows over K , or um rows over $\mathbf{F}_p$, and at most $s + 1$ columns. A loose count for the trace version is $O(m^2u^5(L + 1))$ operations in K . For the direct variant one may bound the cost by

$$O(mu^3L + u^5) \text{ operations in } K, \quad O(m^3u^3 + mu^4) \text{ operations in } \mathbf{F}_p.$$

These counts include powering, exact elimination, dependence tracking, root-product checks and gcds. The prime calls add at most $msP(s, L)$ or $sP(s, L)$ bits respectively. Coefficients remain canonical field coordinates; no coefficient-height growth occurs in elimination. Schoolbook arithmetic and polynomial extended Euclid bound a K operation by $O(m^2(L + 1)^3)$ bits, so these are uniform bit-polynomial reductions. The reference program's simple repeated dependence solves fit within the looser polynomial bounds; operation counters are not a runtime guarantee.

2.3 Multiplicity and inseparability

Normalize f by its leading coefficient. The standard squarefree procedure sets $C = \gcd(f, f')$, $W = f/C$, and $i = 1$. While $W \neq 1$, set $Y = \gcd(W, C)$, emit $(W/Y, i)$ if nonconstant, then set $W = Y$, $C = C/Y$, and increment i . If $C \neq 1$ after this loop, its remaining multiplicities are divisible by p ; its derivative is zero. For $C = \sum c_k X^k$ its unique p th root is

$$H = \sum_j c_{pj}^{p^{m-1}} X^j.$$

Indeed $a^{p^m} = a$ in K . Recurse on H and multiply the returned multiplicities by p . Copying coefficients works only when $m = 1$.

To check this procedure, fix a factor of multiplicity e . If $p \nmid e$, it initially occurs once in W and $e - 1$ times in C , and is emitted at step $i = e$. If $p \mid e$, it remains wholly in C and passes to recursion with multiplicity e/p . This proves the result by induction. The emitted squarefree products have disjoint irreducible supports and sum of degrees at most n . Recursive input degrees sum to at most $2n$. All gcd loops have polynomial cost; coefficient-root exponents have $O(mL)$ bits, not p^{m-1} iterations. Summing the squarefree call bounds gives exactly the bounds in the theorem. The argument covers $p = 2$, $m = 1$ and wholly inseparable inputs.

For checking h , compute successive $T^{p^j} \bmod h$, check the gcd with $T^{p^j} - T$ for every $j \leq \lfloor m/2 \rfloor$, and check $T^{p^m} = T \bmod h$. A reducible degree- m polynomial has an irreducible factor of degree at most $m/2$, so the gcds detect it, also with repeated factors. This uses m binary modular powers and no integer factorization.

3 Prescribed-degree construction

Shoup [2, Theorem 3.1] gives a deterministic reduction from constructing an irreducible degree- N polynomial over $\mathbf{F}_p$ to PF, with polynomial overhead in N, L . Theorem 4.1 gives the analogous arbitrary-base implication. For clarity an alternative known lift is as follows. Construct $H \in \mathbf{F}_p[X]$ irreducible of degree $N = md$, factor it over K by the preceding theorem, and take the lexicographically first factor. The p -Frobenius orbit of a root has size N . p^m -Frobenius acts by adding m modulo N , so every orbit has size $N/\gcd(N, m) = d$. Every factor therefore has degree

d and multiplicity one. This proves the lift without finding an isomorphism or primitive root. It is the same lift as Rai’s Algorithm 2 [3]. For $d = 1$ return X ; for $m = 1$ direct prime construction suffices. The extension is the explicit quotient $K[X]/(g)$ for the output g .

The supplementary construction audit makes Shoup’s oracle use explicit: trial division factors the numeric output degree N ; each prime-power construction has prime-field algebra dimension at most N^2 ; oracle polynomial degrees are at most N^2 , and there are $O(NL \log(N + 1))$ calls. The coefficient slots always have L bits. The bound includes p -power Artin–Schreier towers, odd prime-power binomials and the special 2-power cases. It never factors $p - 1$. Trial division of N is permitted by dense output size and is not an integer-factorization oracle on arbitrary binary inputs. The 2-power exception cannot be omitted: over $\mathbf{F}_3$,

$$X^4 + 1 = (X^2 + X + 2)(X^2 + 2X + 2),$$

although -1 is nonsquare. The supplied construction uses the quadratic extension in that case, following the established binomial criterion. Factoring H over K has input size $O(dm^2L)$, polynomial in the original field and dense degree- d output sizes. Thus the whole construction is bit-polynomial whenever P is polynomial.

4 The upstream input, its audit and the resulting scope

Corollary 2 (Consequence of OpenAI’s prime-field theorem). *Using [4, Theorem 1.1], complete factorization over the represented K and degree- d irreducible construction are uniform deterministic algorithms with bit complexity polynomial in the relevant dense input/output sizes. They require no GRH, randomness, integer-factorization oracle or primitive-root oracle.*

Proof. Substitute the cited prime-field bound $P(u, L) = O(((u + 1)L)^{10^{12}})$ into the two established reductions. Every oracle degree, number of calls and nonoracle representation is bounded by a fixed polynomial in the dense parameters just given. $\square$

This is a consequence of the cited proof, not an independent proof of its analytic or geometric breakthrough. Family 142’s Theorem 1.2 is an auxiliary-prime reduction with numeric prime-size parameter E . Its unconditional Theorem 1.1 imports family 029’s Theorem 1.2 [5]: all finite-order Hecke L -functions over every cyclotomic field containing μ_{12} are zero-free in $\Re s > 1 - 10^{-6}$, with the principal pole permitted. The applications use $F = \mathbf{Q}(\mu_{12q})$ and the cyclic extension $F(p^{1/q})/F$. Abelian zeta factorization and the smoothed explicit formula produce auxiliary primes $\ell \leq cB^{20000000}$, where $B = 20 + (n + 1)(L + 1)$. A Dirichlet-only theorem does not suffice.

We inspected the actual manuscripts at the pinned repository commit

adc7f1241b42e322a6451854ab7e4b4c146bf78a.

The audit covers source hashes, the required theta, reflection, Poisson and Mellin arguments, and the algebraic norm, divisor and splitting proof. Separate automated adversarial audits found no substantive gap in those inspected dependencies. Limited finite checks of auxiliary tables, local Gauss/Euler identities and exact exponent margins are supplied; they do not prove an analytic zero-free theorem. The huge prime-field branch was not executed. Family 003’s actual Lean scope is Dirichlet and Eisenstein-field Hecke nonvanishing; it does not cover the needed varying cyclotomic input. No relevant Lean build was reproduced here and this follow-on is not claimed formally verified.

For optional fixed- r extraction, factor the dense polynomial $X^r - a$ and return its linear-factor roots. Complexity is polynomial in numeric r, m, L , hence in m, L for each fixed r . No polynomial-in- $\log r$ claim follows. This is merely a factoring consequence; quadratic nonresidue and square-root consequences already in the release are not relabeled as new results.

5 Reproduction, limitations and disclosure

The supplement contains exact reduction pseudocode, source/dependency ledgers, proof-node audits and standard-library Python implementations. The bundled prime oracle enumerates $\mathbf{F}_p$ only in bounded small examples; its $O(p)$ cost is not polynomial in $\log p$ and cannot substantiate prime-field derandomization. Independent trial-division comparison checks 377 small monic inputs; a second implementation cross-checks 346 cases. Examples cover characteristic two, nontrivial inverse coefficient Frobenius, repeated factors, $m = 1$, constants and rejected zero/oracle errors. The manuscript makes no assertion of practical efficiency or first priority. The priority audit found only inherited consequences and no demonstrated genuinely new extension. A new-solution Zenodo deposit is therefore withheld.

AI tools were used extensively in research, drafting and verification. Automated adversarial reviews are not conventional human peer review or refereeing; this work has not undergone such review. Alec Kriebel is the sole listed author; no affiliation or outside endorsement is asserted. Upstream AI-produced mathematical work is attributed to OpenAI as supplied by its manuscript-specific citation information.

References

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